WHITE PAPER

Reversible Computing: Preserving Information Through Computation

Recovery, reversibility and their consequences for storage, memory, data transfer and security in the NSATS Quantum portfolio

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Abstract

Many familiar computing operations discard information about their inputs. Retaining that information makes recovery and reuse possible, while requiring memory, processing and management. Recovering an input and carrying out a computation reversibly are different properties: the first concerns information that remains available, the second concerns its preservation through every step. This distinction is central to understanding NSATS's swap gates and the working description “processor-assisted, semi-reversible computing”. Their significance lies in making earlier information available for further use. The consequences extend through storage, working memory, data transfer, security and data centres, where the value of recovery depends on the resources and dependencies that accompany it.

Why information preservation matters

An organisation considering this technology faces three questions:

- What does it gain when earlier inputs remain available for recovery and reuse?
- What does retaining that information require in memory, processing and management?
- How can it distinguish a recoverable result from a reversible computation?

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The answers affect concrete decisions: how much storage to provision, how to size working memory, how to interpret a transfer measurement and what a continuity plan needs to cover. A result alone cannot answer those questions. The information retained with it and the work required to use that information are part of the same computing task.

Context and current status

NSATS licences enable classical devices to perform operations based on quantum mechanics, at room temperature.

Hardware implementations use specially designed classical chipsets without super-cooling.

NSATS sells technology licences, not chips. Its technology helps licensees manufacture their own chips.

Browser and server security is also available as licensed software. White Paper 1, *Applying Quantum Mechanics to Classical Computing*, introduces the portfolio; this paper concentrates on information preservation and its consequences for computation.

NSATS has achieved reversible computing at component level: its swap gates retain input and output information and work reversibly in a limited form. An external processor uses the retained information to regenerate earlier inputs. Full reversibility across the complete system is the subject of continuing research.

Information preservation

An operation is logically reversible when its complete output state uniquely determines its complete input state. “Complete” matters: the result presented to an application may be only part of the information left by the operation. Supporting information retained elsewhere belongs in the description. Landauer's analysis identifies logical irreversibility with the loss of a unique relationship back to the preceding state. [1] For the broader quantum background, White Paper 1 discusses information preservation in the ideal unitary evolution of an isolated quantum system.

Thermodynamic reversibility describes an idealised process with no net entropy production, allowing both the system and its surroundings to be restored to their initial states. Logical invertibility does not, by itself, establish zero practical energy use. [2] This paper uses the logical meaning and returns to energy in the data centre section.

Reversible logic can operate on binary information. Toffoli's work establishes reversible constructions using Boolean variables. [3] The distinction between reversibility and the number of signal states matters later, because NSATS's quantum-classical bus uses more than two states.

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What a result can lose

An AND operation returns 1 only when both inputs are 1. The other input combinations all return 0. The table shows this standard logical example, with 0 meaning false and 1 meaning true.

Input A	Input B	AND result alone	Result with both inputs retained
0	0	0	0, with inputs 0 and 0
0	1	0	0, with inputs 0 and 1
1	0	0	0, with inputs 1 and 0
1	1	1	1, with inputs 1 and 1

The result alone merges three different input pairs. Retaining both inputs preserves their identity: the complete output is then the result together with both inputs. Auxiliary information plays this role in embedding a function reversibly. [3]

The example demonstrates preservation of distinguishable inputs, not compression. Preserving distinctions does not require every encoded item to keep the same length, a point the storage section returns to.

Recovery, history and the complete computation

As a general illustration, consider a calculation whose result loses distinctions between inputs, while a separate copy of the input is kept. The earlier input can be recovered from the copy. That demonstrates recovery from a record; it does not show whether every step of the complete operation, including the handling of that record, preserves the information needed to identify its preceding state.

Recoverability concerns whether an earlier input can be obtained from the information available. Logical reversibility requires each step within the chosen boundary to preserve enough information to identify the state before it. A recoverable transition may also be reversible, but recovery alone does not establish that all the steps meet this test. [1] [4] This determines which retained information belongs in the storage and memory accounts, and where the energy argument about avoiding erasure applies.

Bennett's construction shows one way to preserve information through an entire computation. It executes the forward computation reversibly while recording a history, copies the result, then runs that reversible computation backwards to clear the history. The backward pass depends on the forward steps having been executed reversibly. An arbitrary irreversible computation cannot generally be run backwards. [4]

History recording can consume more storage as a computation proceeds; other reversible operations, such as permutations, need not accumulate a history. Overwriting is logically irreversible when it discards distinctions from the complete state. A register can change without losing those distinctions. The method used therefore affects both processing time and storage requirements, as Bennett's analysis of reversible simulation demonstrates. [1] [5]

For an implementation, the boundary can include gates, retained information, the controller, processor and storage, together with transfer where applicable. A smaller subsystem can be considered if its inputs, outputs and external dependencies are explicit. The boundary defines the steps being tested and the resources being counted. The next section applies this test to NSATS's gates.

NSATS's swap gates

NSATS's swap gates remember data that has passed through them. The retained information allows an earlier input to be regenerated and the corresponding output to be generated. Earlier information remains available for recovery and reuse in subsequent calculations.

For any gate that retains information, its state before an operation includes both the arriving input and the information already retained. Its state afterwards includes the output and the updated retained information. The reversibility test covers that complete change.

In established logic, the abstract SWAP operation exchanges its inputs. The controlled-SWAP, or Fredkin gate, makes that exchange conditional on a control input. These abstract operations preserve input information in their outputs and need no history of earlier operations. [6] NSATS's swap gates differ from these abstract operations in the respect relevant here: they retain information from earlier operations.

Processor-assisted, semi-reversible computing

NSATS uses “processor-assisted, semi-reversible computing” to describe computation in which its gates retain input and output information and an external processor uses that information to regenerate an earlier input or generate the corresponding output.

“Semi-reversible” indicates that reversibility is achieved at component level, in the gates, while full reversibility of the complete computation, including the processor's work, is the subject of continuing research. It is an NSATS working description, not an established scientific category or a measured degree of reversibility. The presence of a processor does not by itself make a computation irreversible; what matters is how information is preserved through its operations.

Gate research

Research into gate upgrades covers forward and reverse operation, the management of retained information and the role of the external processor. The test described above applies to any resulting arrangement: each step within its defined boundary has to preserve the information needed to identify the preceding state.

Signals and the quantum-classical bus

Reversibility concerns the information an operation keeps; the bus concerns how signals represent and carry information.

NSATS's quantum-mechanical logic in classical devices supports compression, encryption, transfer and processing, working with a processor. The quantum-classical bus carries electrical signals with more than two states, beyond binary 0 and 1. The bus enables faster processing, data transfer and decision making within a quantum-classical electronic device.

Within finite, effectively specified digital representations and transitions, neither additional signal levels nor reversibility changes which functions are computable. Turing's computational model and Bennett's reversible simulation provide the relevant background. [7] [4]

A symbol's information capacity depends on how many alternatives can be distinguished reliably. Noise constrains that distinction. Hartley and Shannon describe these theoretical limits on representation and communication. [8] [9]

Temporal Compression and storage

Temporal Compression allows more original data to be held within the same physical storage resource, reducing the number of drives needed for the same data. The gain comes from storing fewer bits for the original data; the physical storage units retain their existing bit capacity. NSATS's processor-assisted, semi-reversible computing contributes to compression, so the drive holds more data.

The AND example preserved information by retaining the inputs. Storage introduces a different question: how few bits can represent original data while keeping it retrievable intact? Physical capacity describes the underlying resource; effective capacity describes how much original data can be stored and retrieved intact through its encoded representation. White Paper 3, *Temporal Compression: Understanding Effective Storage Capacity*, examines these distinctions in depth. This paper considers what retrieval requires.

Effective capacity depends on processing time, not only physical size. A temporal drive is better described by how much data it can process and store per second than by a fixed capacity.

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An infrastructure decision needs both the rate at which data is stored and retrieved and the amount retained for retrieval.

For existing storage, the licensee adds a specially designed chipset or controller built with the licensed technology. The device remains ordinary classical hardware. This applies to hard drives, solid-state drives, USB flash drives, optical storage and other storage types. Continuity planning includes the added component and any retained information needed for retrieval, with required dependencies distinguished from replaceable components.

Repeated compression, in which a compressed output passes through further compression, is part of the licensed technology. Further storage methods are the subject of continuing research.

Lossless compression exploits redundancy. Already compressed or securely encrypted material often leaves little exploitable redundancy, although neither label establishes that a particular item is incompressible. No lossless method shortens every possible input, and repeated passes do not change this limit. [10]

Memory and RAM, Processors and Internal Buses

Classical data encoded using quantum-mechanical logic can be sent and retrieved faster over computer buses, giving faster access to working information in existing RAM.

Existing RAM always participates through a chipset or controller that the licensee adds, built with the licensed technology. The RAM itself remains ordinary classical hardware. Memory-controller designers have a role alongside RAM manufacturers because access to that working information depends on the controller as well as the memory.

Gates retain information, processors perform operations, controllers manage access, RAM supplies working information, and internal buses carry data between components. Research continues into further processing methods.

Working RAM and retained storage have different roles in the resource account, which also includes retained gate information and the processor's working memory. Bennett's analysis addresses general time and space trade-offs in reversible simulation. [5]

Quantum-Classical Networking

Quantum-classical network cards at both endpoints enable high-speed data transfer while the routers between them remain ordinary. The technology serves wired, optical and wireless links, including long-distance connections. Research also covers further transfer methods.

The useful outcome is original data available to the receiving application. Several different measures describe the path to that outcome.

Measure	What it counts	Unit	What it tells a decision maker
Channel capacity	The maximum rate of reliable information transmission under a channel model	Bits of information per second	The information-rate constraint imposed by the channel
Transmitted-bit rate	Bits sent on a particular link per unit time	Bits per second	The rate of the representation sent over that link
Original-data throughput	Original data delivered per unit time	Original bytes per second, or original bits per second	The rate of delivery of usable original data
Elapsed time to usable data	Time from the start of the transfer task until the receiving application can use the data, including endpoint work	Seconds	How long the complete transfer takes

Original-data throughput depends on both channel capacity and the source's information content per original byte. Counted in bits per second, original data can be reconstructed faster than bits are transmitted where the original contains redundancy. Channel capacity still constrains the achievable rate for a given source. Shannon relates the rate of original symbols to source entropy and channel capacity. [9] [10]

Post-Quantum Security

Retaining earlier inputs also retains information that needs protection. The licensee's security assessment includes that information alongside the current inputs and results.

NSATS's post-quantum security uses classical computing methods inspired by quantum mechanics. For browsers and the servers they connect to, NSATS licenses the security as software under standard software licence terms. For larger systems, licensees build the security into their own chips.

Recovering information and protecting it impose different requirements. A reversible transformation preserves the ability to recover an input; confidentiality concerns who can obtain it. Invertibility alone provides no secrecy. Protection depends on the cryptographic construction, appropriate key handling and the threat against which it is evaluated. Shannon's analysis of secrecy systems provides the scientific background. [11]

Protection also extends to the handling of retained information at the end of use. The implementation's applicable sanitisation arrangements belong in that review; NIST's media sanitisation guidance addresses this final stage. [12]

Data centres

With the licensed technology, fewer hard drives and less memory are needed to store data, existing memory processes existing binary data faster, and the chipsets run with lower heat than conventional computing.

For data centre, high-performance computing and hosting providers, the practical test is the net resource requirement. Drive and memory reductions are considered alongside the gates, retained information, controller and processor resources needed by the implementation.

The theoretical energy argument concerns erasure. Avoiding logical erasure removes erasure's theoretical contribution to dissipation, while practical energy use includes other operations and components. A system combining reversible and irreversible steps can reduce erasure while incurring costs elsewhere. Logical reversibility alone does not guarantee a net energy saving. [1] [2]

Chipset heat, server power and facility cooling are separate measures; a change in one does not determine the others.

Applications and licensed intellectual property

The applications span temporal storage, RAM and memory controllers, processors and buses, networking equipment, security and data centre systems. Storage, network and memory manufacturers address component and controller functions. Data centre, high-performance computing and hosting providers consider the complete workload, while browser and application providers address the software security offering.

NSATS licenses its intellectual property: patents, designs and copyrighted software. Where the licensed IP takes the form of software, NSATS supplies it under the licence, with its documentation and updates. Each licensee designs and builds its own solution.

A licensing enquiry can be framed around the intended application, required functions and implementation.

The executive decision

Recovery and reuse are valuable when earlier information serves the organisation's workload. Their practical requirements can be evaluated alongside the licensed storage, memory, transfer and security functions. White Paper 1's assessment protocol provides the general approach to evaluating a complete task.

The evidence for that decision includes the original and stored sizes in bits, all information needed for intact retrieval, and the processing used to store and retrieve it. Working RAM is accounted for separately from retained storage, with gate information and processor working memory included according to their roles. Transfer records distinguish the measures in the networking table. Energy records follow the separation set out in the data-centre section. The resource account includes the added components needed by the implementation.

Four groups of questions bring these requirements together:

- **Information:** what is retained, where, for how long, and what can be regenerated under the intended operating conditions?
- **Computation:** what boundary is being considered, and what does the processor contribute within it?
- **Resources:** how do processing and memory requirements vary with workload and duration?
- **Continuity and use:** what does retrieval depend on, which components can be replaced, and how does the implementation serve the intended workload?

The licensed capabilities can be assessed against those questions now. Continuing research addresses gate operation, retained information and further storage, processing and transfer methods. The distinction running through both is precise: recovery makes earlier information available again; reversible computation preserves the information needed to identify the preceding state at every step within the defined boundary.

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Scientific background references

These works provide the scientific background to the principles discussed.

1. Landauer, R. (1961). *Irreversibility and Heat Generation in the Computing Process*. IBM Journal of Research and Development, 5(3), 183–191. DOI: 10.1147/rd.53.0183.
2. Bennett, C. H. (1982). *The thermodynamics of computation—a review*. International Journal of Theoretical Physics, 21, 905–940. DOI: 10.1007/BF02084158.
3. Toffoli, T. (1980). *Reversible computing*. Automata, Languages and Programming, Lecture Notes in Computer Science, 85, 632–644. DOI: 10.1007/3-540-10003-2_104.
4. Bennett, C. H. (1973). *Logical Reversibility of Computation*. IBM Journal of Research and Development, 17(6), 525–532. DOI: 10.1147/rd.176.0525.
5. Bennett, C. H. (1989). *Time/Space Trade-Offs for Reversible Computation*. SIAM Journal on Computing, 18(4), 766–776. DOI: 10.1137/0218053.
6. Fredkin, E. and Toffoli, T. (1982). *Conservative logic*. International Journal of Theoretical Physics, 21, 219–253. DOI: 10.1007/BF01857727.
7. Turing, A. M. (1936–37). *On Computable Numbers, with an Application to the Entscheidungsproblem*. Proceedings of the London Mathematical Society, s2-42(1), 230–265; publisher record dated 1937. DOI: 10.1112/plms/s2-42.1.230.
8. Hartley, R. V. L. (1928). *Transmission of Information*. Bell System Technical Journal, 7(3), 535–563. DOI: 10.1002/j.1538-7305.1928.tb01236.x.
9. Shannon, C. E. (1949). *Communication in the Presence of Noise*. Proceedings of the IRE, 37(1), 10–21, especially section XI, Theorem 4 and the following discussion. DOI: 10.1109/JRPROC.1949.232969.
10. Shannon, C. E. (1948). *A Mathematical Theory of Communication*. Bell System Technical Journal, 27(3), 379–423; 27(4), 623–656. DOI, first instalment: 10.1002/j.1538-7305.1948.tb01338.x; DOI, second instalment: 10.1002/j.1538-7305.1948.tb00917.x.
11. Shannon, C. E. (1949). *Communication Theory of Secrecy Systems*. Bell System Technical Journal, 28(4), 656–715. DOI: 10.1002/j.1538-7305.1949.tb00928.x.
12. Chandramouli, R. and Hibbard, E. (2025). *Guidelines for Media Sanitization*. NIST Special Publication 800-88 Rev. 2. DOI: 10.6028/NIST.SP.800-88r2.

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